



Technical Service Bulletin

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Engine Will Not Crank or Cranks Slowly (Electric Starter): Debris and Dust in Electric Starting Motor

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK23 CM500
- QSK23 CM500 K157G
- QSK23 CM2250 K109

Issue Summary

Symptom:

- Engine will **not** crank
- Engine cranks slowly

Root Cause:

- Nose bushing of electric starting motor contamination causing nose bushing and drive end bushing wear.

Verification

Visual inspection of electric starting motor can confirm if dust and debris has collected in nose cone of electric starting motor where slow cranking or no cranking of the engine is experienced.

Resolution

- If dust and debris has collected in the nose cone of the electric starting motor, a noseless electric starting motor option is available. See Table 1 for part number information.
- For proper wiring and connection, reference Service Instructions section below.

Service Instructions

- For more information on the M128 starting motor and installation, reference the Prestolite™ TSBs found in Table 2.

Note : Changes in OEM starter wiring, and connector lug sizes may be required when upfitting from the Komatsu® starting motor to the new M128 starting motor. See Table 1 for M128 lug sizes.

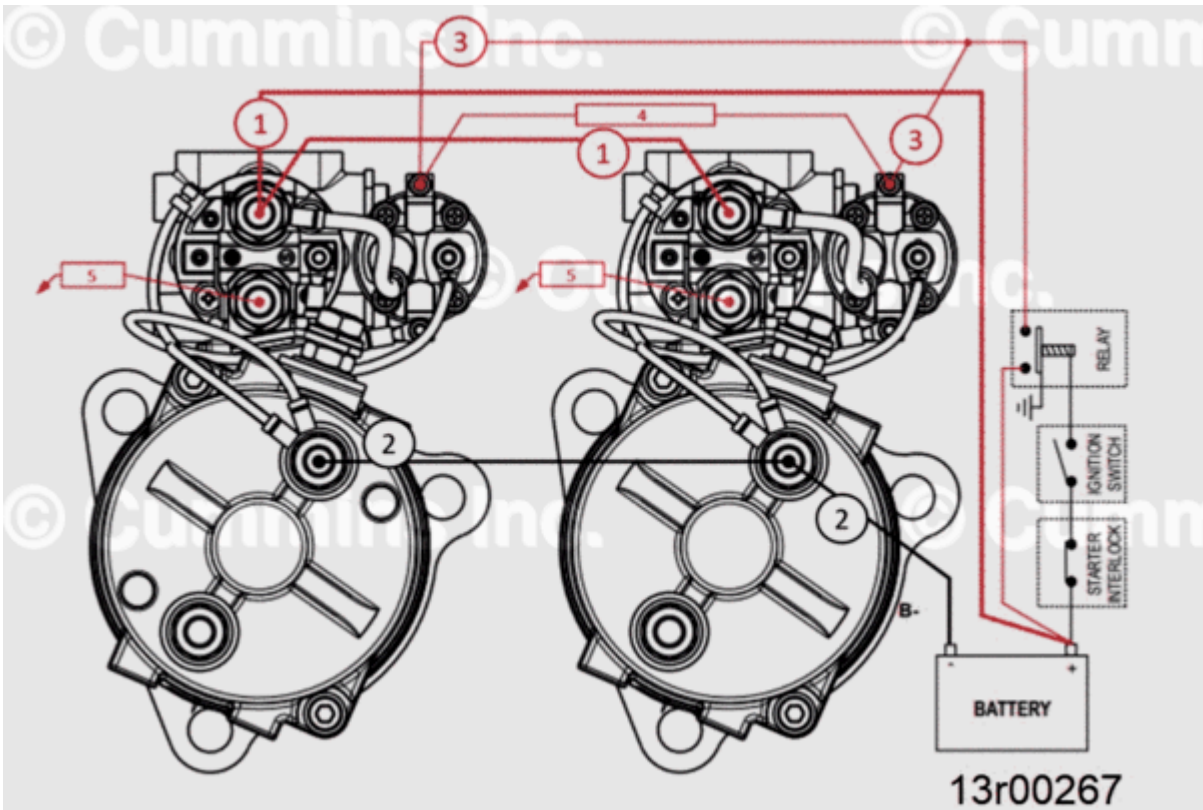


Figure 1, Dual-Starting Motor M128 Wiring Diagram.

Table 1, Figure 1 Wiring Diagram Callouts

Callout	Description	Lug Size
1	BATT. +	M12 x 1.75
2	BATT. -	M12 x 1.75
3	IGNITION	M5 x 0.8
4	Balance Lead *	M5 x 0.8
5	Sensing Lead **	M12 x 1.75

* Install an ignition input balancing wire connecting the two starter ignition input terminals. Wire size **must** be at least 16 awg.

** Some dual start applications have voltage sense functionality. This is specific to special customer applications and starters. Ignore if system is not equipped with this feature. Prestolite™ recommends the customer connects a sensing terminal so that the operator can observe if both starters have achieved meshing. If no lead is present, this may void warranty. Contact technical service if unsure.

Table 2, Related Prestolite™ Publications	
Prestolite Description	TSB Number
24V M128R Starters - Wiring / Terminal Identification	TSB-1171
M128 – Dual Starter Wiring Procedure	TSB-1172
MS7 – M128 Supersession Fitment Information	TSB 4079

Service Parts

Service parts are offered. See Table 3 for part numbers.

Table 3, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Motor, Starting	4396011	No	No	5593352

Part Compatibility

- When upfitting dual-starter installations, both starting motors **must** be the new available part number. Mismatch starting motors are **not** supported.
- **Never** remove or bypass the integrated magnetic switch (IMS) on the M128 starting motor. This integrated starting relay is part of the starter soft start feature.

Document History

Date	Details
2024-2-19	Module Created